Question 1 (20 marks)

Solve the following problems. To apply any elementary row operation to a matrix, you should specify the elementary row operation using the standard notations professor introduced during the lecture.

(a) Solve the following system of linear equations using augmented matrix and describe the solutions in **parametric vector form** or explain why the system has no solution.

$$\begin{cases} x_1 + 3x_2 + x_3 = 1 \\ 4x_1 + 9x_2 - 2x_3 = 1 \\ 3x_2 + 6x_3 = 3 \end{cases}$$

[5 marks]

(b) Compute the determinant of

$$B = \begin{bmatrix} -3 & 7 & 2 & 9 \\ 1 & -4 & 2 & 3 \\ 0 & 1 & -2 & 1 \\ 3 & -2 & 3 & 4 \end{bmatrix}.$$

[5 marks]

(c) Find the inverse of

$$C = \begin{bmatrix} 1 & -2 & 1 \\ 5 & -7 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

[5 marks]

(d) Let D be a 3×3 matrix with the property that $D^3 = 0$, show that the matrix $D + 2I$ is **invertible** by finding its inverse. Here

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is the 3×3 matrix. Verify that the matrix in your answer is really the inverse of $D + 2I$.

[5 marks]

Question 2 (16 marks)

(a) Let T_1 be a linear transformation that sends

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \text{ to } \begin{pmatrix} 2 \\ 5 \end{pmatrix}, \text{ and } \mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \text{ to } \begin{pmatrix} -3 \\ 4 \end{pmatrix}.$$

Find the image of $2\mathbf{u} + 3\mathbf{v}$ under T_1.

[4 marks]

(b) Let $T_2 : \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation given by

$$T_2(\mathbf{v}) = \begin{bmatrix} 1 & 4 & -5 \\ 3 & -7 & 4 \end{bmatrix} \mathbf{v}.$$

Determine (i) if T_3 is one to one, and (ii) if it is onto. Justify your answers.To apply any elementary row operation to a matrix, you should specify the elementary row operation using the standard notations professor introduced during the lecture.

[6 marks]

(c) Determine if the following sets of vectors are **linearly independent** or **linearly dependent**. Justify your answers. If you want apply an elementary row operation to a matrix, you should specify the elementary row operation using the standard notations professor introduced during the lecture.

(i) $$\left\{ \begin{bmatrix} -4 \\ 0 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \\ -4 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 3 \\ 6 \end{bmatrix} \right\}$$

(ii) $$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

[6 marks]

Question 3 (14 marks)

(a) Let $\mathbf{v}_1 = \begin{bmatrix} 2 \\ -8 \\ 6 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 4 \\ 6 \\ -7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 6 \\ -10 \\ 11 \end{bmatrix}$, and $E = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix}$.

Determine if $\mathbf{u}$ is in Col E, the column space of E. To apply any elementary row operation to a matrix, you should specify the elementary row operation using the standard notations professor introduced during the lecture. [5 marks]

(b) Given a matrix F and an echelon form G of F as follows

$$F = \begin{bmatrix} 1 & 3 & -1 & 3 \\ -2 & 1 & 5 & 4 \\ -1 & 2 & 1 & 2 \\ 0 & -2 & 3 & 1 \end{bmatrix} \sim G = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 7 & 3 & 10 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(i) Find a basis for Row F, the row space of F. [3 marks]

(ii) Find a basis for Col F, the column space of F. [3 marks]

(iii) Find a basis for Nul F, the Null space of F. [3 marks]

Question 4 (10 marks)

(a) Let $\mathcal{A} = \{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be bases for a vector space V. Suppose $\mathbf{b}_1 = 2\mathbf{a}_1 - \mathbf{a}_2 + 4\mathbf{a}_3$, $\mathbf{b}_2 = \mathbf{a}_1 - \mathbf{a}_2 + 3\mathbf{a}_3$, and $\mathbf{b}_3 = 5\mathbf{a}_1 - 2\mathbf{a}_3$. Let $\mathbf{x} = 3\mathbf{b}_1 - 2\mathbf{b}_2 + \mathbf{b}_3$.

(i) Find $[\mathbf{x}]_{\mathcal{B}}$. [2 marks]

(ii) Find $[\mathbf{x}]_{\mathcal{A}}$. [3 marks]

(b) Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$ be bases for a vector space V. Assume the coordinate vector $[\mathbf{x}]_{\mathcal{C}}$ of $\mathbf{x}$ with respect to $\mathcal{C}$ is given by $[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} -2 \\ -1 \\ -5 \end{bmatrix}$ and the change-of-coordinates matrix $P_{\mathcal{C}\leftarrow\mathcal{B}}$ is given by

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 0 \\ 2 & -1 & -3 \end{bmatrix}.$$

Find the coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ with respect to the basis $\mathcal{B}$. [5 marks]

Question 5 (20 marks)

Assume $\{\mathbf{x}_1, \mathbf{x}_2\} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} \right\}$ is a basis for a subspace W. Let $A = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

(a) Use the Gram-Schmidt process to produce an orthogonal basis for W. [5 marks]

(b) Find the least square solution of $A\mathbf{y} = \mathbf{b}$. [5 marks]

(c) Construct a singular value decomposition of the A. [10 marks]

Question 6 (20 marks)

(a) Let A and B be two $n \times n$ matrices such that $AB = BA$. Assume that $\mathbf{v}$ is an eigenvector of A corresponding to the eigenvalue $\lambda = 2$. Prove that $B\mathbf{v}$ is an eigenvector of A. What is the corresponding eigenvalue? [5 marks]

(b) Prove that if A is an orthogonal matrix, namely, $A^T A = I$, then any real eigenvalue of A must be either 1 or -1. [5 marks]

(c) Consider the quadratic form

$$f(x_1, x_2, x_3) = 3(a_1x_1 + a_2x_2 + a_3x_3)^2 - 4(b_1x_1 + b_2x_2 + b_3x_3)^2.$$

(i) Let $\boldsymbol{\alpha} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Show that the matrix of the quadratic form is

$$3\boldsymbol{\alpha}\boldsymbol{\alpha}^T - 4\boldsymbol{\beta}\boldsymbol{\beta}^T.$$

[5 marks]

(ii) Show that if $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are unit vectors and are orthogonal to each other, then there is an orthogonal change of variable

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = P \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},$$

so that $f(x_1, x_2, x_3)$ can be written as $3y_1^2 - 4y_2^2$. [5 marks]

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